

ACKNOWLEDGEMENTS

The publisher would like to thank the following people for their help with making the book: Agnibesh Das, John Gillespie, Anita Kakkar, Sophie Parkes, Antara Raghavan, and Rupa Rao for editorial assistance; Revati Anand and Priyanka Bansal for design assistance; Vishal Bhatia for CTS assistance; Jeffrey E Post, Ph D Chairman, Department of Mineral Sciences Curator, National Gem and Mineral Collection, National Museum of Natural History, Smithsonian; Kealy Gordon and Ellen Nanney from the Smithsonian Institution; Ruth O'Rourke for proofreading; Elizabeth Wise for indexing; and RGB Research Ltd (periodictable.co.uk), especially Dr Max Whitby (Project Director), Dr Fiona Barclay (Business Development), Dr Ivan Timokhin (Senior Chemist), and Michal Miškolci (Production Chemist).

The publisher would like to thank the following for their kind permission to reproduce their photographs:

(Key: a-above; b-below/bottom; c-centre; f-far; l-left; r-right; t-top)

9 Bridgeman Images: Golestan Palace Library, Tehran, Iran (cra). **Fotolia:** Malbert (ca). **Getty Images:** Gallo Images Roots Rf Collection / Clinton Friedman (fcla). **Wellcome Images** <http://creativecommons.org/licenses/by/4.0/>: (cra). **13 Getty Images:** Stockbyte (cra). **Science Photo Library:** McGill University, Rutherford Museum / Emilio Segre Visual Archives / American Institute Of Physics (cra). **15 Science Photo Library:** Sputnik (cra). **17 Alamy Stock Photo:** Dennis H. Dame (cr). **20 Dreamstime.com:** Alek79 (cb). **NASA:** X-ray: NASA / CXU / Univ Potsdam / L.Osknova et al; Optical: NASA / STScI; Infrared: NASA / JPL-Caltech (cra). **21 Alamy Stock Photo:** Phil Degginger (cb); ULA (fcr). **NASA:** Bill Rodman (cra). **Science Photo Library:** U.S. Navy (cra). **24 Alamy Stock Photo:** PjStudio (cra). **Dreamstime.com:** Titovstudio (ca). **naturepl.com:** Christophe Courteau (cr). **25 123RF.com:** Federico Cimino (cr). **Dreamstime.com:** Aleksey Boldin (cla); Polygomaki (ca). **Getty Images:** Corbis (cla/mirror); Driendl (cra). **NASA:** (cra). **Science Photo Library:** 26 **123RF.com:** Stellargems (cra); Sara Winter (cra). **Dorling Kindersley:** Tim Parmenter / Natural History Museum, London (cr). **27 123RF.com:** Todsaporn Bunmuen (cl); Francis Dean (cra). **Alamy Stock Photo:** Artspace (cr); Hemis (cl). **Dreamstime.com:** Abel Tumik (clb). **28-29 Alamy Stock Photo:** Hemis. **30 Alamy Stock Photo:** Siim Sepp (c). **31 123RF.com:** Petkov (ca). **Alamy Stock Photo:** Doug Steley B (cb). **Dorling Kindersley:** Dave King / The Science Museum, London (clb). **Dreamstime.com:** Mohammed Anwarul Kabir Choudhury (cr); Jarp3 (cra). **Getty Images:** John B. Carnet (cl). **Alamy Stock Photo:** CLAIRE PAXTON & JACQUI FARROW (cl). **32 123RF.com:** Dario Lo Presti (cra). **Getty Images:** De Agostini Picture Library (cr). **33 123RF.com:** Lenise Calleja (c/cracker); Chaiyaphong Kitphaphaisan (cr). **Alamy Stock Photo:** David J. Green (c). **Dreamstime.com:** Robert Semmic (cl). **Getty Images:** Stocktrek Images (cla). **Natural Resources Canada, Geological Survey of Canada:** (cb). **34 Dorling Kindersley:** Oxford University Museum of Natural History (cla). **Getty Images:** Ullstein Bild (clb); Universal Images Group (cra). **35 Alamy Stock Photo:** Universal Images Group North America LLC / DeAgostini (cla). **Getty Images:** Keystone-France (cla). **39 123RF.com:** Vladimir Kramin (cra). **Alamy Stock Photo:** Craig Wise (cla). **Dreamstime.com:** Studio306 (cla/sprinkler). **Getty Images:** iStock Images - Caspar Benson (cb). **NASA:** NASA / MSFC / David Higginbotham (c). **Science Photo Library:** David Parker (cr). **Wellcome Images** <http://creativecommons.org/licenses/by/4.0/>: Wellcome Library (cra). **40 Dorling Kindersley:** Colin Keates / Natural History Museum, London (cla). **41 123RF.com:** Thodonal (cb). **Alamy Stock Photo:** Mohammed Anwarul Kabir Choudhury (clb/cement); Dominic Harrison (cla); Phil Degginger (clb). **Dreamstime.com:** Nui1983 (cr); Marek Uliasz (cra). **Getty Images:** Yoshikazu Tsuno (cra). **Rex by Shutterstock:** Neil Godwin / Future Publishing (cl). **42 Alamy Stock Photo:** Phil Degginger (cl). **Dorling Kindersley:** Natural History Museum, London (clb); Holts Gerns (cla). **43 123RF.com:** Oksana Tkachuk (c). **Alamy Stock Photo:** Ekasit Wangprasert (cb). **Dreamstime.com:** Waxart (cr). **44-45 Alamy Stock Photo:** Inge Johnsson. **47 123RF.com:** Anatol Aduskevich (cra); Pawel Szczepafski (ca); Ronstik (cra). **Dorling Kindersley:** Durham University Oriental Museum (cla). **Dreamstime.com:** Showface (cr). **iStockphoto.com:** Lamiel (cl). **48-49 Alamy Stock Photo:** The Natural History Museum (cb). **49 123RF.com:** Roman Ivashchenko (cr); Wieslaw Jarek (ca). **Getty Images:** DEA / S. VANNINI (c). **Science Photo Library:** ALAIN POL, ISM (cra). **51 Getty Images:** Heritage Images (cra). **Science Photo Library:** Public Health England (ca, cra); Public Health England (cb). **54 123RF.com:** Stocksnapper (cb). **Alamy Stock Photo:** Universal Images Group North America LLC / DeAgostini (cra). **Dreamstime.com:** Dimitar Marinov (cra). **55 123RF.com:** Leonid Pilnik (fcr); Sergei Zhukov (cr). **Alamy Stock Photo:** Military Images (cra); Hugh Threlfall (clb). **Dreamstime.com:** Flynt (cra). **56 123RF.com:** Mykola Davydenko (fcr); Kaetana (cra). **Alamy Stock Photo:** Shawn Hempel (cl). **57 Alamy Stock Photo:** imageBROKER (cra). **Dorling Kindersley:** Natural History Museum, London (cr). **58 Alamy Stock Photo:** Vincent Ledvina (clb). **59 123RF.com:** Chaiyaphong Kitphaphaisan (cra); lightboxx (c); Tawat Langnamthip (cra). **Alamy Stock Photo:** Hemis (cb); B.A.E. Inc. (ca). **Dreamstime.com:** Nexus7 (cr). **Getty Images:** Michael Nicholson (cra). **60 123RF.com:** Sereznily (clb). **Getty Images:** Detlev van Ravenswaay (cl). **61 Alamy Stock Photo:** PhotoCuisine RF (c); SERDAR (f). **Dorling Kindersley:** Doubleday Holbeach Depot (cra). **Dreamstime.com:** Igor Sokolov (clb). **Science Photo Library:** Jim West (cr). **62-63 123RF.com:** Wang Aizhong. **64 Alamy Stock Photo:** Susan E. Degginger (cra); The Natural History Museum (cr). **65 Dorling Kindersley:** Rolls Royce Heritage Trust (cl). **Dreamstime.com:** Margogh (c). **Getty Images:** Pascal Preti (cb); Science & Society Picture Library (cla). **66 Alamy Stock Photo:** Alan Curtis / LGPL (cla). **67 123RF.com:** Psvrsuo (ca); Евгений Косцов (cra). **Alamy Stock Photo:** INTERFOTO (fcla). **Dorling Kindersley:** National Music Museum (cla). **Getty Images:** Panthomme Hubert (cra). **68 Alamy Stock Photo:** Jeff Rotman (cra). **Dorling Kindersley:** Natural History Museum, London (ca); Oxford University Museum of Natural History (clb). **69 123RF.com:** Dilyana Kruseva (cra); Vitaliy Kityayko (cla); Photopix (cra). **Alamy Stock Photo:** Paul Ridsdale Pictures (tc). **Dorling Kindersley:** University of Pennsylvania Museum of Archaeology and Anthropology (cb). **70-71 Alamy Stock Photo:**

Novarc Images. **72 Alamy Stock Photo:** Phil Degginger (cla). **73 Alamy Stock Photo:** PjStudio (clb). **Dreamstime.com:** Sean Pavone (cra). **NASA.** **74 Dorling Kindersley:** Oxford University Museum of Natural History (cla). **75 123RF.com:** Belchonok (cr); Weerayos Surarengchai (cla). **Alamy Stock Photo:** dpa picture alliance (ca); Georgios Kollidas (cra); PNWL (cl). **Getty Images:** SSPL (cb). **Science Photo Library:** David Parker (cra). **76 123RF.com:** Okan Akdeniz (clb); Nevarpp (clb). Andriy Popov (cra). **Dreamstime.com:** Ryan Stevenson (cb). **77 123RF.com:** Mohammed Anwarul Kabir Choudhury (cra); Vladimir Nenov (cra). **Alamy Stock Photo:** The Natural History Museum (cl). **NASA.** **78 Alamy Stock Photo:** Oleksandr Chub (cra); The Natural History Museum (ca). **Science Photo Library.** **78-79 Alamy Stock Photo:** Susan E. Degginger (c); epa european pressphoto agency b.v. (cb). **79 Science Photo Library:** David Parker (cra); Rvi Medical Physics, Newcastle / Simon Fraser (cr). **80 123RF.com:** Missisiya (cb); Darren Pullman (clb). **Alamy Stock Photo:** GFC Collection (cra). **81 123RF.com:** Hywit Dimyadi (cra). **Dreamstime.com:** Shutterstock99 (ca). **Getty Images:** Alain Nogues (cra). **82 Alamy Stock Photo:** Greenshoots Communications (ca); PjStudio (cra). **Dreamstime.com:** Robert Chlopas (cr). **Science Photo Library.** **83 123RF.com:** Dmitry Lobanov (cra); Jose Ignacio Soto (tr); Valerii Zan (cra). **Dreamstime.com:** Maloy40 (cra). **Getty Images:** Paul Taylor (cl). **84 Getty Images:** DEA / PHOTO 1 (clb); DEA / G.CIGOLINI (cl). **85 Alamy Stock Photo:** David J. Green (cra); Chromorange / Juergen Wiesler (cra). **Dorling Kindersley:** The University of Aberdeen (ca). **Dreamstime.com:** Stephanie Frey (cla); Gaurav Masand (cl). **Getty Images:** Science & Society Picture Library (clb). **Science Photo Library.** **86 123RF.com:** Serhii Kucher (cra). **Alamy Stock Photo:** Ableimages (cra/micro). **Dreamstime.com:** Michal Baranski (cra). **Getty Images:** Lester V. Bergman (cra). **Science Photo Library.** **87 Dreamstime.com:** Andrew Eremin (clb). **Science Photo Library.** **88 123RF.com:** Ludinko (cra). **Getty Images:** Trisha Leeper (cra). **89 123RF.com:** Akulamatia (cra); Anton Starikov (clb). **Dreamstime.com:** Homydesign (cb). **90 Alamy Stock Photo:** Antony Nettle (cra). **Dreamstime.com:** Farbled (c); Vesna Njagulj (clb). **91 Alamy Stock Photo:** Science (cra). **Dreamstime.com:** Reddicks (cra). **Science Photo Library:** Dr Gopal Murti (ca); Dirk Wiersma (cla). **92 Alamy Stock Photo:** Citizen of the Planet (cra). **Getty Images:** Yva Momatuk and John Eastcott (cra). **Science Photo Library.** **93 123RF.com:** Sergey Jarochkin (cra); mg154 (cl). **Alamy Stock Photo:** Pictorial Press Ltd (cl). **NASA:** CXU / NGST (ca). **94 Dorling Kindersley:** Natural History Museum, London (r). **Science Photo Library:** Natural History Museum, London (l). **95 Alamy Stock Photo:** Four sided triangle (cl); iStudio (ca); Friedrich Saurer (cra). **Dreamstime.com:** Adamanto (cr). **Getty Images:** PHAS (cra); Royal Photographic Society (cl). **Science Photo Library:** Dr P. Marazzi (fcr); National Physical Laboratory (C) Crown Copyright (cla); Sovereign / Ism (cra). **96 Science Photo Library:** Science Stock Photography (cra). **97 123RF.com:** Ratchaphon Chaihuai (clb). **Dorling Kindersley:** Alistair Duncan / Cairo Museum (ca); Barnabas Kindersley (cl). **Dreamstime.com:** Nastya81 (cra); Scanrail (c). **Getty Images:** Charles O'Rear (fcr); John Phillips (cr). **magiccarpics.co.uk:** Robert George (cb). **NASA.** **98-99 Alamy Stock Photo:** imageBROKER. **101 123RF.com:** Teerawut Masawat (cla). **Getty Images:** Science & Society Picture Library (ca); Science & Society Picture Library (cr). **Paul Hickson, The University of British Columbia:** (cl). **102 Getty Images:** Bettmann (cla). **Science Photo Library:** Ernest Orlando Lawrence Berkeley National Laboratory / Emilio Segre Visual Archives / American Institute Of Physics (clb). **103 Alamy Stock Photo:** Peter van Evert (cra); Randsc (clb). **Science Photo Library:** Lawrence Berkeley National Laboratory (cr). **104 Science Photo Library:** David Parker (clb); Wheeler Collection / American Institute Of Physics (cla); David Parker (cb). **105 Alamy Stock Photo:** imageBROKER (cla). **Science Photo Library:** Emilio Segre Visual Archives / American Institute Of Physics (cl). **106 Alamy Stock Photo:** Granger Historical Picture Archive (clb). **Science Photo Library:** David Parker (cla). **107 Alamy Stock Photo:** Sherab (cl). **Science Photo Library:** Dung Vo Trung / Look At Sciences (cr). **110 123RF.com:** Oleksandr Marynchenko (clb); Naruedom Yaempongsak (cra). **Alamy Stock Photo:** John Cancelli (cla); Reuters (cb). **111 123RF.com:** Cobalt (cra); Veniamin Kraskov (cra/red). **Dreamstime.com:** Akulamatia (cb). **Science Photo Library.** **112 Alamy Stock Photo:** Everett Collection Historical (cra). **113 Alamy Stock Photo:** Ivan Vdovin (cra). **Fotolia:** Efrid (cra). **114 Rex by Shutterstock:** (cra). **Science Photo Library:** Pr Michel Brauner, ISM (cra). **115 Alamy Stock Photo:** G M Thomas (cr). **Science Photo Library:** Patrick Llewellyn-Davies (cra). **116 123RF.com:** Vereshchagin Dmitry (br); Vitalii Tiahunov (cra); Vitalii Tiahunov (fcr). **117 123RF.com:** Preecha Bamrungrai (cra). **Dreamstime.com:** Hxdzbzy (cra). **120 Alamy Stock Photo:** Yon Marsh (br). **ESA:** (cra). **Science Photo Library:** Dirk Wiersma (clb). **121 Alamy Stock Photo:** Mike Greenslade (f). **Science Photo Library:** Trevor Clifford Photography (cra); Sputnik (cra). **122 Alamy Stock Photo:** Derrick Alderman (cr). **Science Photo Library:** J.C. REVY, ISM (cra); Lawrence Berkeley Laboratory (cra). **123 NASA:** NASA / JPL-Caltech / Malin Space Science Systems (cra). **Science Photo Library:** Theodore Gray, Visuals Unlimited (ca). **124 Alamy Stock Photo:** Randsc (cra). **Dreamstime.com:** Marcorubino (clb). **NASA.** **Science Photo Library:** Science Source (cla). **125 Alamy Stock Photo:** 501 collection (cla). **Getty Images:** George Rinhart (clb). **Science Photo Library:** US Department Of Energy (ca). **126 Science Photo Library:** American Institute Of Physics (cra); Sputnik (cb); Sputnik (clb). **127 Alamy Stock Photo:** Granger Historical Picture Archive (clb). **Science Photo Library:** Ernest Orlando Lawrence Berkeley National Laboratory / Emilio Segre Visual Archives / American Institute Of Phys (cla). **130 123RF.com:** Terry Davis (cla). **Alamy Stock Photo:** Chris stock photography (fcr). **130-131 Alamy Stock Photo:** Universal Images Group North America LLC / DeAgostini (cb). **131 123RF.com:** Sirichai Asawalapsakul (ca); Joerg Hackemann (cla); Wilawan Khasawong (ca/boric); Michal Giel (c/TV). **Alamy Stock Photo:** Chronicle (fcr). **Dorling Kindersley:** Tank Museum (cra). **Fotolia:** L. amica (c); Alex Staroseltsev (cr). **Getty Images:** Heritage Images (cra). **132 Science Photo Library:** Dirk Wiersma (ca). **133 123RF.com:** Destinacigdem (cla); Olaf Schulz (cb). **Dreamstime.com:** Apple Watch Edition™ is a trademark of Apple Inc., registered in the U.S. and other countries. (cr); Stepan Popov (cl); Simon Gurney (fcr); Zalakdagli (clb). **134-135 Getty Images:** Brasil2. **136 123RF.com:** Martin Lehmann (c). **Alamy Stock**

Photo: BSIP SA (cra). **Getty Images:** Visuals Unlimited, Inc. / GIPhotoStock (cra/disc). **NASA.** **137 123RF.com:** Norasit Kaewsai (cb/trans); Ouhdesire (clb); Dmytro Sukharevsky (cb). **Dreamstime.com:** Christian Delbert (cra). **138 Dreamstime.com:** Monika Wisniewska (cl). **Getty Images:** Science & Society Picture Library (cr). **139 123RF.com:** Fotana (clb). **Alamy Stock Photo:** Stock Connection Blue (cla). **Getty Images:** The Asahi Shimbun (cr). **142 Alamy Stock Photo:** Pablo Paul (cra); WidStock (cla). **Dorling Kindersley:** Natural History Museum (cra); Natural History Museum (fcr). **143 123RF.com:** Oleksii Sergiev (cra). **Alamy Stock Photo:** David J. Green (cla); Image.com (ca). **Dorling Kindersley:** National Cycle Collection (ca/cycle); The Science Museum, London (fcr). **144-145 Bridgeman Images:** Christie's Images. **146 123RF.com:** Danilo Forcellini (fcr). **Alamy Stock Photo:** Phil Degginger (cra); Perry van Munster (cra). **147 123RF.com:** Scanrail (ca). **Alamy Stock Photo:** MixPix (cla); Haiyin Wang (cra). **Dreamstime.com:** Halil I. Inci (fcr). **Getty Images:** Handout (cr). **Science Photo Library:** Lawrence Berkeley National Laboratory (cb). **148 123RF.com:** Viktoriya Chursina (cra). **Dreamstime.com:** Bright (cra); Oleksandr Lysenko (cr). **Getty Images:** DEA / G. CIGOLINI (cla). **149 123RF.com:** Lapis2380 (clb). **150 Alamy Stock Photo:** Sarah Brooksby (cb). **Dorling Kindersley:** Natural History Museum, London (cla). **151 123RF.com:** Vira Dobosh (clb). **Science Photo Library:** Sputnik (cr); Sputnik (cb). **154 Science Photo Library:** Dr.Jeremy Burgess (clb). **155 123RF.com:** Mohammed Anwarul Kabir Choudhury (cra/color); Teerawut Masawat (cla); David Gilbert (cla). **Alamy Stock Photo:** Lyroky (cra); Tim Scrivener (cra). **NASA:** JPL (cra). **156-157 Getty Images:** Icon Sports Wire. **158 Dorling Kindersley:** Natural History Museum, London (cr/beaker). **Dreamstime.com:** Tomas Pavelka (c). **Getty Images:** Auscape (ca). **158-159 Science Photo Library.** **159 123RF.com:** Action sports (cb); De2marco (c); Arina Zaichin (clb); Mohammed Anwarul Kabir Choudhury (cra). **Getty Images:** Simone Brandt (cr). **160 123RF.com:** Maksym Yemelyanov (cra). **Alamy Stock Photo:** Andrew Ammendolia (fclb). **Dreamstime.com:** Jagat (clb); Science Pics (ca). **161 Dreamstime.com:** Chris Boswell (cra). **Getty Images:** SuperStock (cra). **Science Photo Library:** Phil Degginger (cla); Phil Degginger (cla). **162 Dorling Kindersley:** Harry Taylor (cla). **163 123RF.com:** Sereznily (cl). **Dreamstime.com:** Ericlefrancais (cla); Bert Folsom (clb). **Science Photo Library:** Sputnik (cr). **166 123RF.com:** Kameel (clb); Russ McElroy (cra). **167 123RF.com:** Rostislav Ageev (cra/diver). **Alamy Stock Photo:** PhotoAlto (cra); RGB Ventures / SuperStock (fcr). **Dreamstime.com:** Narin Phapnam (cb); Uatp1 (cl). **Getty Images:** STR (clb). **SuperStock:** Cultura Limited / Cultura Limited (cra). **168 123RF.com:** Cseh Ioan (clb). **Alamy Stock Photo:** Big Pants Productions (cb). **Science Photo Library:** Farrell Grehan (cra). **168-169 Alamy Stock Photo:** The Natural History Museum (ca). **169 123RF.com:** Lucian Milasan (cra); Nikkytok (c). **Alamy Stock Photo:** Krys Bailey (clb); Paul Felix Photography (cra). **Dreamstime.com:** Nfransua (ca); Kirsty Pargeter (cla); Olha Rohulya (cb). **170-171 Getty Images:** Kazuyoshi Nomachi. **172 123RF.com:** Maksym Bondarchuk (cra); Sautetas (cb). **Dreamstime.com:** Orjinal (clb). **Science Photo Library:** Dirk Wiersma (cra). **173 123RF.com:** Jiri Vlacelk (ca). **Alamy Stock Photo:** Hugh Threlfall (fcr); Universal Images Group North America LLC / DeAgostini (cla). **Getty Images:** Steve Proehl (cra). **174 Alamy Stock Photo:** Dan Leeth (clb). **175 Alamy Stock Photo:** CPC Collection (cb); Sputnik (cla). **178 Dorling Kindersley:** Natural History Museum, London (cra); Oxford University Museum of Natural History (cla). **179 123RF.com:** Kiril Krasnov (c); Chaovarut Shoop (clb). **Alamy Stock Photo:** The Print Collector (cra); World History Archive (cl); World foto (cr). **Dreamstime.com:** Bogdan Dumitru (fcr); Stephan Pietzko (cla). **Getty Images:** John B. Carnet (cra). **181 123RF.com:** Sergey Jarochkin (c); Dmitry Naumov (cr); Hxdzbzy (cb/bleach). **Alamy Stock Photo:** Maksym Yemelyanov (cra). **Dorling Kindersley:** Thackeray Medical Museum (cl). **182-183 Science Photo Library:** Alexis Rosenfeld. **184-185 Getty Images:** George Steinmetz (cb). **Science Photo Library:** Charles D. Winters (c). **185 Dreamstime.com:** Jose Manuel Gelpi Diaz (cra); Larry Finn (ca). **Science Photo Library.** **186 123RF.com:** Alexandr Malyshev (fcr); r fñ (cra). **Alamy Stock Photo:** BSIP SA (cra). **187 Science Photo Library:** Union Carbide Corporation's Nuclear Division, courtesy EMILIO SEGRE VISUAL ARCHIVES, Physics Today Collection / AMERICAN INSTITUTE OF PHYSICS (clb). **190-191 Dreamstime.com:** Andrey Navrotsky (b). **190 123RF.com:** Leonid Ikan (cb). © CERN: (cra). **191 Dreamstime.com:** Yinan Zhang (cb). **iStockphoto.com:** Gogbigpicture (t). **Science Photo Library:** Brian Bell (ca); Patrick Landmann (cra). **192-193 Getty Images:** Rolf Geissinger / Stocktrek Images. **194 123RF.com:** Rainer Albiez (cra). **Alamy Stock Photo:** D. Hurst (cra). **Science Photo Library:** Andrew Lambert Photography (clb). **195 Dreamstime.com:** Stocksolutions (cra). **Getty Images:** Floris Leeuwenberg (cra); Mario Tama (clb). **Science Photo Library:** Crown Copyright / Health & Safety Laboratory (cra). **196 Dorling Kindersley:** Clive Streeter / The Science Museum, London (cra). **Dreamstime.com:** Liouthe (cra). **Getty Images:** Gemya Savilov (fcr). **Science Photo Library:** (clb). **197 123RF.com:** Alexlmm (fcr). **Alamy Stock Photo:** Alexandru Nika (cra). **Dreamstime.com:** Jultud (clb). **Getty Images:** Brand X Pictures (cra). **NASA:** JPL-Caltech (clb). **198 Science Photo Library:** Dirk Wiersma (ca). **199 123RF.com:** Nmint (cl). **Alamy Stock Photo:** ITAR-TASS Photo Agency (cra); Gordon Mills (cla); RGB Ventures / SuperStock (clb); ITAR-TASS Photo Agency (cra). **200 Alamy Stock Photo:** Shawn Hempel (bc). **205 Dorling Kindersley:** Natural History Museum (cla)

All other images © Dorling Kindersley
For further information see: www.dkimages.com